Introduction to Bonding

Friday, 10 July 2020 1:31 PM

Covalent

- Covalent Bond: Strong electrostatic forces of attraction between a shared pair of electrons and the nuclei of atoms on either side
- May be simple molecular ($Cl_{2(g)}$, $CH_{4(g)}$, $H_2O_{(l)}$) OR covalent network (diamond, graphite, SiO_2)

Properties - Simple Molecular

- Non-Conductors: Localised electrons within each atom/covalent bond
- Soft/Weak: Only weak intermolecular forces between molecules
- Low Melting/Boiling Point: Only the weak intermolecular forces must be overcome, not the strong intramolecular forces

Properties - Covalent Network

- Non-Conductors: No mobile particles
- Hard/Brittle: Strong covalent bonds between all atoms
- Very High Melting/Boiling Point: Lots of energy required to break covalent bonds
- *Graphite is an exception to the first two

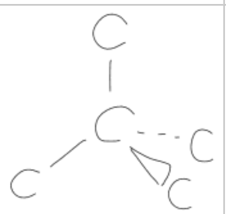
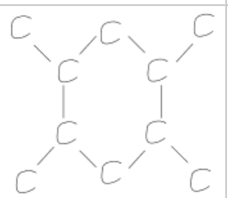
Predicting Molecular Geometry (VSEPR)

a. Draw dot structure showing valence electrons
    - Least electronegative (excluding H) is the central atom
    - All atoms (but especially central) should have an **octet**
b. Count electron clouds around central atom
c. Predict geometry of electron clouds around central atom
d. Ignore lone pairs and state shape

Bonding Pairs	Lone Pairs	Shape	Angle	Example
2	0	Linear	180	CO_2
3	0	Trigonal Planar	120	SO_3
3	1	Bent	119	SO_2
4	0	Tetrahedral	109.5	CH_4
4	1	Pyramidal	107	NH_3
4	2	Bent	104.5	H_2O

Carbon Allotropes

- Allotropes: Physical forms an element can take

Allotrope	Structure	Properties
Diamond		• 4 covalent bonds per atom (tetrahedral) • Highest melting/boiling point known • Hard, brittle, non-conductor
Graphite		• 3 covalent bonds per atom (trigonal planar) • One delocalised e^- per atom, holds layers together • Soft, conductor, *lower* melting/boiling point (still high)

Ionic

- Ionic Bond: Strong electrostatic forces of attraction between oppositely charged ions in a network
- Ions held in 3D lattice structure where each ion has 6 oppositely charged ions surrounding it
- Nitrates, ethanoates, ammonium salts and group 1 salts are almost always soluble

Properties

  - Non-conductor (solid): No mobile charged particles
  - Conductor (liquid/aqueous): Mobile charged ions
  - High melting/boiling point: Lots of energy to overcome strong electrostatic forces
  - Hard but brittle: Very strong bonds (hard); Small displacement causes contact between like ions and they repel each other (brittle)

Hydrated vs Anhydrous

  - Hydrated: Contains water molecules as part of its crystalline structure
  - Anhydrous: Does not contain water molecules
  - Water of crystallisation: Water molecules that make up part of an ionic crystal

Metallic

- Metallic Bond: Electrostatic forces of attraction between a sea of delocalised electrons and metal cations in a network
  - Metal atoms lose electrons forming stable cations and a sea of delocalised electrons
- Cations are in fixed positions, only electrons move

Properties

  - Electrical Conductors: Mobile charged particles (electrons)
  - Thermal Conductors: When heated, electrons vibrate (kinetic energy) and knock into each other as they move; chain reaction along entire metal
  - Malleable/Ductile: Non-directional bonds; ions can move without breaking them
  - High Melting/Boiling Point: Lots of energy to overcome strong electrostatic forces

Intermolecular Forces

Thursday, 30 April 2020 11:10 AM

Intermolecular vs Intramolecular Forces

- *Intra*molecular force are the strong electrostatic forces of attraction holding atoms within a molecule together
- *Inter*molecular forces are the weak electrostatic forces of attraction holding molecules together
    - 10-100 times weaker than covalent bonds
- When simple molecular substances are melted/boiled, ONLY intermolecular forces are broken
    - E.g. Water -> Steam: Molecules close together in water as there are intermolecular forces of attraction, heating water breaks these bonds and forms steam, still H_2O (covalent bonds not broken)

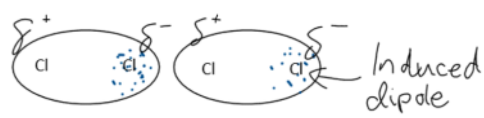
Polarity of Molecules

- Have to consider:
    - Electronegativity of each atom
    - Shape of molecule (symmetry)
- If atoms are different, the more electronegative atom develops a slightly negative charge
    - "Dipole bond" if electronegativity difference > 0.4
- If there are only 2 atoms in a molecule with a dipole bond, the molecule is polar
- If there are multiple dipole bonds, but the molecule is symmetrical it is non-polar
- E.g. CO_2 is linear with angle 180 degrees (symmetrical), therefore non-polar molecule although it has 2 dipole bonds, as they cancel each other out

Types of Intermolecular Forces

- 3 types
    - Dispersion
    - Permanent dipole-dipole
    - Hydrogen bonding

Dispersion

- Occurs in all substances
- Dispersion forces are the temporary, fluctuating dipoles created due to the random movement of electrons (protons don't move)
- Larger molecules have stronger dispersion forces as there are more electrons to create a stronger dipole
- Linear molecules have stronger dispersion forces as there is more surface area to contact other molecules
- E.g. Cl_2 is non polar, therefore only dispersion forces occur between Cl_2 molecules
    - The random movement of electrons may result in most of the 34 electrons to be on one side of the molecule at any point in time, creating an *induced dipole* on neighbouring molecules as like charges repel and opposites attract
    - 

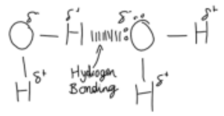
Permanent Dipole-Dipole

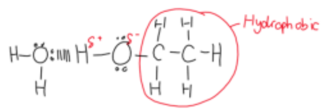
- Polar molecules can attract each other to form permanent dipole-dipole forces

- Larger dipole = stronger bond
- Stronger than dispersion forces as it is not temporary or fluctuating

Hydrogen Bonding

- Hydrogen Bond: The strong dipole-dipole attraction between an electron deficient hydrogen atom attached to an electronegative atom (N, O or F) and the lone pair of electrons on a highly electronegative atom attached to another hydrogen atom on a different molecule
  - I.e. A hydrogen atom bonded to an N, O or F atom, bonded to a lone pair of electrons on an N, O or F atom bonded to another H atom on another molecule
  - E.g. H_2O, C_2H_6O, NH_3, HF
- Hydrogen bonds are the strongest intermolecular force as the dipole is so large
- Molecules with a larger electronegativity difference will have stronger hydrogen bonds
  - E.g. H_2O has stronger hydrogen bonds than NH_3

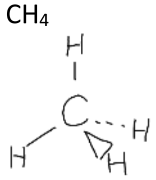
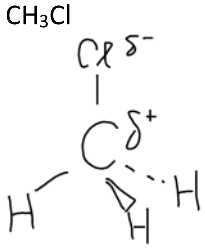




Intermolecular Forces and Physical Properties

- Increased melting/boiling point from:
  - Increased number of electrons (molecular mass)
  - Permanent dipole-dipole bonds (vs dispersion)
  - Hydrogen bonds
- Vapour pressure (tendency to evaporate)
  - Weaker intermolecular forces = high vapour pressure
  - Higher temperature = higher vapour pressure
- "Like dissolves like"
  - Polar solutes will generally dissolve in polar solvents and non-polar solutes in non-polar solvents
    Non-Polar solute won't dissolve in polar solvents as they can only form dispersion forces which can't overcome the dipole-dipole forces in the solvent

VSEPR/IMF Examples

Molecule	Intermolecular forces	Bonding Pairs	Lone Pairs	Shape	Angle
CH_4 	Dispersion	4	0	Tetrahedral	109.5
CH_3Cl 	Dispersion Permanent dipole-dipole	4	0	Tetrahedral	109.5

Chromatography

Friday, 1 May 2020 1:58 PM

Overview

- All forms of chromatography have a mobile phase which moves through or across a stationary phase
  - Stationary Phase:
    - Compounds in the mixture are adsorbed (attracted) and slowed down
    - Either solid or liquid
  - Mobile Phase:
    - The more soluble components in the mixture are carried faster as the mobile phase moves
    - Either liquid or gas
- Separation depends on components' solubility in the mobile phase compared to their tendency to adsorb to the stationary phase
- **Retention Factor (R_f)** = distance travelled by solute / distance travelled by solvent
  - It will always be between 0 and 1
- **Retention time:** The amount of time passed between injecting the sample to the detector recording it

Paper Chromatography (PC)

- **Stationary Phase:** *Liquid molecules* (often water) bonded to chromatography paper
- **Mobile Phase:** Liquid solvent

  ### How it works
  - Small amount of the unknown is dissolved and spotted onto a **pencil** line at one end of the chromatography paper
  - Mobile phase moves up stationary phase through capillary action
  - More polar components will adsorb to the stationary component as it is also very polar, while those with a greater tendency to remain dissolved in the mobile phase will move further

Thin Layer Chromatography (TLC)

- **Stationary Phase:** Layer of silica gel fixed onto a glass plate (polar)
- **Mobile Phase:** Liquid solvent
- Works the same way as paper chromatography
- Lid placed on the top of the beaker
  - Closed system, allows solvent to saturate atmosphere & keep dynamic equilibrium
- If some substances in the mixture don't separate using one solvent, flip the plate 90 degrees and do it again with a different solvent (remember to mark the starting position with a pencil)

Capillary Action

- Capillary Effect: A combination of adhesive and cohesive forces which allows the mobile phase to move through/across the stationary phase
  - Adhesive forces are between molecules of **different substances**; enables solvent molecules to bond to the stationary phase and not fall down
  - Cohesive forces are between molecules of the **same substance**; enables solvent molecules to bond to each other and move each other through the stationary phase

Gas Chromatography (GC)

- **Stationary Phase:** High-boiling-point/Non-volatile liquid coating inside of column (e.g. silica)
- **Mobile Phase:** Inert gas
- Not suitable for compounds that decompose before vapourising when heated (generally

molecular weight should be < 300g per mol)
- Components separate based on **volatility**; more volatile components will have a greater tendency to stay in the MP and will move faster (lower retention time)

How it works

- Stationary phase packed into a stainless steel column (2-5mm diameter 1.5-10m length)
- Liquid sample injected into injecting chamber and **vaporised**
- Mobile phase sweeps the sample through the column
- The column is in an oven and set at a temperature to produce a good separation of the components
  - Temperature may be set to gradually increase over time so the less volatile components won't take so long to exit the column
- Detector measures the amount of a substance at any given time and plots it onto a graph (**gas chromatogram**)
- **Flame Ionisation Detector (FID)** is a commonly used detector, gives retention time but doesn't identify the substance
- **Mass Spectrometer (MS)** can be used as the detector to identify substances as they exit the column - **Gas Chromatography Mass Spectrometry (GCMS)**

High Performance Liquid Chromatography (HPLC)

- **Stationary Phase:** Fine solid particles (often silica gel)
  - Small particles are essential as they present a **larger surface area** of SP that the sample can interact with
- **Mobile Phase:** Liquid solvent (usually a mixture of water/acetonitrile/methanol)
  - Sample must be soluble in MP
- Column usually 3-5mm diameter and 10-30cm length
- Suitable for samples with a high molecular mass
- How it works
  - Column tightly packed with the SP
  - MP passed under high pressure from a pump through the column containing SP
  - Separation based on **polarity**
    - **Normal Phase HPLC:** SP polar, MP non-polar; polar analytes longer retention time
    - **Reverse Phase HPLC:** SP non-polar, MP polar; polar analytes shorter retention time

Identifying Analytes and Concentrations from a Chromatogram

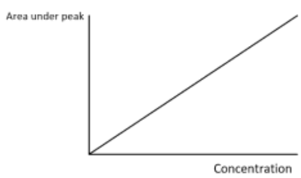
- Peaks characterised by **height**, **area** and **retention time**
  - These can be used to **identify the analytes** and **calculate their concentration**

Identifying Analytes - Retention Time

- Chromatogram of the pure substance being tested for produced under identical conditions
- Retention time of the pure substance is compared to the retention times of the peaks of the sample being tested
- If the retention times match, the analyte has been identified

Concentration - Area under the peak

- **Concentration is directly proportional to the area under the peak**
- To determine the concentration of an analyte, we want to obtain a graph of **concentration vs area under the peak**



- To get this graph, a set of **standards** are run (chromatographs for known concentrations of the substance) under the exact same conditions and the graph is produced (line of best fit)
- The area under the peak of the unknown substance is calculated and plotted on the

graph to obtain the concentration

Applications of HPLC and GC

- Both ideal for determining identity and concentration of analytes within an unknown sample
    - **Reliable**, **efficient**, **fast** and **non-destructive**
    - Only uses tiny samples
- GC requires sample to be vapourised, so if it decomposes before vapourising it cannot be applied
    - These compounds are usually separated using HPLC
- HPLC requires sample to be soluble in a suitable solvent and its polarity allows for separation

GC Applications

- Detecting dangerous gases in underground mines
- Environmental pollutant monitoring
- Forensic analysis
- Testing for illegal/banned substances in sports
- Analysis of refined products (e.g. Petroleum)

HPLC Applications

- Quality control and analysis of products (oils, vitamins, sugars, caffeine, medicine, fats, antioxidants)
- Analysis of amino acids, peptides and proteins in biochemical research

In Class

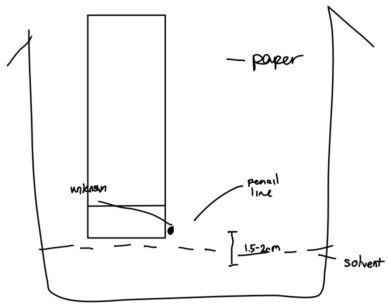
Wednesday, 6 May 2020 2:59 PM

What is Chromatography?

- Chromatography is a series of analytical techniques that can be sued to separate mixtures of compounds for further use or for analysis
- I.e. It is a general term for many different separation techniques
- All forms of chromatography have a mobile phase which moves through or across a stationary phase
  - Stationary Phase:
    - Compounds in the mixture are attracted to it (*ad*sorbed) and slowed down
    - Either solid or liquid
  - Mobile Phase:
    - The more soluble components in the mixture are carried faster as the mobile phase moves
    - Either liquid or gas
- Retention Factor (Rf) = distance travelled by solute / distance travelled by solvent
  - It will always be between 0 and 1

Paper Chromatography

- Used to separate mixtures, particularly dyes or pigments
- Stationary phase is the paper
- Mobile phase is the liquid (usually water)
- Key points:
  - Unknown mixture placed on a PENCIL line (pen ink will affect the chromatography)
  - Solvent in beaker can't touch the mixture (it will dissolve)



Thin Layer Chromatography (TLC)

- The stationary phase is a layer of silica gel fixed onto a glass plate
- Mobile phase is a solvent which travels up the plate, carrying the substances
- Separation depends on solubility or charge
- If some substances in the mixture don't separate using one solvent, flip the plate 90 degrees and do it again with a different solvent (remember to mark the starting position with a pencil)
- Things we might be asked:
  - Which component is more polar? (with diagram) - component which travelled less distance is more polar as it is more attracted to the silica gel

Capillary effect

- The solvent (mobile phase) is drawn up the stationary phase by capillary action
  - Adhesive forces are between molecules of different substances; enables solvent molecules to bond to the stationary phase
  - Cohesive forces are between molecules of the same substance; enables solvent molecules to bond to each other and move each other through the stationary phase
- The capillary effect is a combination of both of these forces and is how the mobile phase 'climbs' up the stationary phase

TLC vs Paper Chromatography

- TLC has many advantages over paper chromatography
  1. The glass plate is rigid unlike paper so it is easy to control
  2. After separation, the components of the mixture can be collected; the silica gel holding the separated substances is scraped off the glass plate and added to a solvent; the substance will dissolve and the silica gel can be easily removed by filtration
  3. Environmentally friendly

UV and Locating Agents (TLC)

- Many substances are white or colourless, so aren't visible on a TLC plate
- One way of making colourless substance show up is to use UV light
  - Usually works well for organic compounds
- Another method is to use a chemical locating agent (a chemical that reacts with the substance to produce a coloured compound)
  - E.g. When ninhydrin is exposed to an organic compound it stains is purple-brown

Column Chromatography

- **Stationary Phase:** Small beads of silica gel (SiO_2) or alumina (Al_2O_3)
- **Mobile Phase:** Liquid solvent

How it works

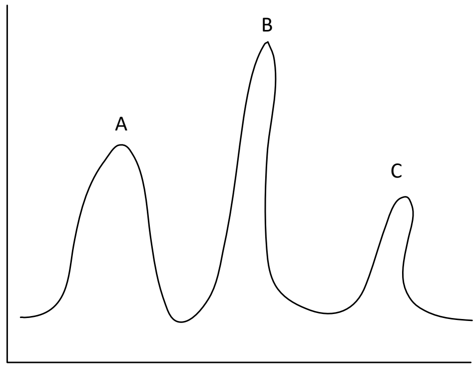
- Tube packed with stationary phase and filled to top with mobile phase
- Sample mixture placed on top
- Stopcock opened and gravity drags mobile phase through the stationary phase; It drips out the bottom
- More mobile phase is added to the top of the column at the same rate it exits the column
- Components of the substance will be dragged down at different speeds and can be

collected in separate beakers by opening and closing the stopcock

Gas Chromatography (GC)

- Used in many analytical laboratories such as forensic police labs, synthetic chemical labs and drug testing labs
- Stationary phase usually a long thin tube of silica gel (liquid)
- Mobile phase is gas (must be inert - unreactive)
- Sample injected into a machine and **vaporised**, then washed over with an inert gas (MP)
- **Some substances are more attracted to the SP**, they will take longer to reach the detector
- Higher boiling point will result in being more attracted to SP as it is closer to its liquid phase than the lower BP substances, but higher boiling point can be a result of polar molecules, so still dependent on polarity
- Detector measures the amount of a substance at any given time and plots it onto a graph called a **gas chromatogram**

Gas Chromatogram



Retention time

Testing for banned substances

- Racehorses are regularly tested for banned substances such as painkillers that help them run through injury, or steroids
- Urine samples are collected from the horses at events and sent to labs to be tested by gas chromatography

High Performance Liquid Chromatography (HPLC)

- MP: Liquid (usually a mixture of water/acetonitrile/methanol)
- MP passed under pressure through column containing SP
- Sample must be soluble in MP
- Suitable for samples with a high molecular weight
- Column is tightly packed with silica gel particles

Lucarelli Ch 19

Wednesday, 6 May 2020 8:10 AM

19.1 - Chromatography : An Introduction

- Instrumental chromatographs can analyse microlitre (10^{-6}L) amounts of a substance
- The can detect picogram (10^{-12}g) amounts of a substance in the samples
- Separation of components depends on solubility (which depend on intermolecular forces) in the mobile phase compared to their tendency to adsorb (stick to the surface of) to the stationary phase

19.2 - Paper Chromatography

- Stationary Phase: Water molecules bound to chromatography paper
- Mobile Phase: Liquid solvent
- A small amount of the unknown is dissolved and spotted on a pencil line at one end of the chromatography paper
- Mobile phase moves up stationary phase through capillary action
- More polar components will adsorb to the stationary component as it is also very polar, while those with a greater tendency to remain dissolved in the mobile phase will move further
- If the components are colourless they can still be identified by using UV light or a locating agent (ninhydrin is very common)

19.3 - Column Chromatography

- Stationary Phase: Small beads of silica gel (SiO_2) or alumina (Al_2O_3)
- Mobile Phase: Liquid solvent
- Glass column is packed with the stationary phase and filled up with the mobile phase
- A small amount of the unknown substance is placed on top
- The stopcock is opened and the mobile phase will start dripping out of the column
- More solvent is added at the same rate it is flowing out
- The components will separate based on their polarity (more polar will adsorb to the stationary phase and move slower, less polar will dissolve in solvent better and move faster)

19.4 - Thin Layer Chromatography

- Stationary Phase: Glass plate coated with silica (SiO_2) or alumina (Al_2O_3)
- Mobile Phase: Liquid solvent
- Mobile phase moves up stationary phase through capillary action
- Mixture will separate based on polarity (more polar will adsorb to the stationary phase and less polar will dissolve better in the mobile phase)
- Produces better separation than paper chromatography
- Retention Factor (R_f) is used to identify substances
    - $R_f = \frac{distance\ moved\ by\ solute}{distance\ moved\ by\ solvent}$
- The R_f of a substance is dependent on the stationary and mobile phase as well as the temperature
    - To determine the substances, a TLC of the pure substances must be conducted under identical conditions
- Simple and inexpensive; used for separating inks, dyes, plant extracts and drug testing
    - If positive for drug testing, a confirmatory test is needed as it sometimes gives false positives

19.5 - Instrumental Chromatography

- Chromatography techniques in research and industry usually use techniques such as Gas Chromatography (GC) and High Performance Liquid Chromatography (HPLC)
- The principles of separation for these are the same as the simpler methods
    - Separation of the analytes (components of the mixture) depends on their attraction to the stationary vs mobile phase

19.6 - Gas Chromatography

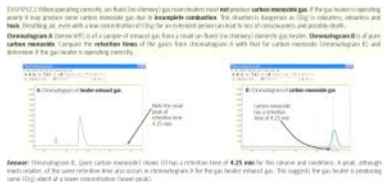
- **Stationary Phase:** High-boiling-point/non-volatile viscous liquid adsorbed onto solid particles such as silica
- **Mobile Phase:** Inert carrier gas
- Ideal for identifying individual substances and concentrations in mixtures
- *Not suitable for compounds that decompose when heated or don't vaporise easily (molecular*

mass above 300g per mol)

How it works

- The stationary phase is packed into a **stainless steel column** (usually 2-5mm diameter and 1.5-10m length)
    - Capillary columns are less than a millimetre in diameter and up to 100m length
- The liquid sample is **injected** into an injecting chamber via a microsyringe
    - The chamber is up to 250 degrees, and vapourises the sample which then flows with the MP into the column
- The column is in an **oven** and is set at a temperature to produce good separation of the components
- The different components will separate based on **volatility**; More volatile will have a greater tendency to stay in the MP and have a lower retention time, and vice versa
- As they reach the end of the column they are analysed by a detector which records them on a chromatogram
- **Retention time of a substance is the amount of time passed from injecting the sample to the detector recording a it passing out the end**
- It depends on **BP/volatility** as well as **column temperature**
    - A 10 degree increase in column temperature will approximately halve the retention time
    - The column temperature sometimes slowly increases so that the less volatile substances won't take so long to exit the column
- The **Flame Ionisation Detector (FID)** is a common detector used to detect substances exiting the column - gives retention time but doesn't identify the substance
- A Mass Spectrometer (MS) can be used as the detector to identify the substances
    - Known as **Gas Chromatography Mass Spectrometer (GCMS)**

19.7 - Identifying Analytes from Retention Time



- Peaks are characterised by **height**, **area** and **retention time**
- These features can be used to **identify analytes** and **determine their concentration**

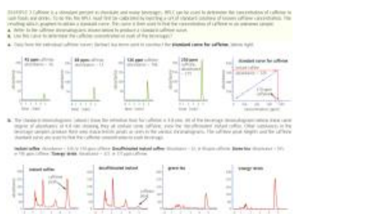
How it works

- A chromatogram of the sample of the pure substance being tested for must be produced under the **exact same conditions** (temp, gas and gas flow rate, column)
- The retention time of the pure substance is **compared** to the retention times of the unknown sample being tested

19.8 - High Performance Liquid Chromatography (HPLC)

- **Stationary Phase:** Fine solid particles (e.g. silica) packed into a column
- **Mobile Phase:** Liquid solvent
- If gas chromatography cannot be used for a particular sample, HPLC is usually used instead
- The column is usually 3-5mm diameter and 10-30cm length
- Particle size is very important as it presents a **larger surface area** for the analytes to **interact** with
- Separation based on **polarity**
    - In **Normal Phase HPLC** the SP is polar and MP is non-polar, so more polar analytes will have a longer retention time
    - In **Reverse Phase HPLC** (more common) the SP is non-polar and MP is polar, so more polar analytes will have a shorter retention time
    - Relative polarity of the SP and MP can be altered by slowly changing their composition over time so that analytes which aren't moving much will be released and flow out with the MP
- An **Ultraviolet Light Detector** is often used to detect the presence of analytes in the **eluent** (solution exiting the column)
    - Requires that the analytes absorb UV light

19.9 - Concentration Analysis from HPLC Chromatogram



- To determine the concentration of a substance, the chromatograph must be calibrated to it first
    - A **set of chromatographs** for **known concentrations** of the substance are created, and a graph of **absorbance vs concentration** is obtained from the data
- The **same chromatograph and conditions** are used on the unknown sample, and the absorbance of the corresponding retention time can be **plotted on the graph** obtained earlier to calculate the concentration

19.10 - Applications of GC and HPLC

- Both GC and HPLC are ideal for determining identity and concentration of substances within an unknown sample

    - **Reliable**, **efficient**, relatively **fast** and **non-destructive (components can be collected)**
    - Requires only minute samples
- GC requires the sample to be in a **vaporised state** so can only be applied to substances which **vapourise before decomposing** (generally molar **mass < 300g per mol**)
- Compounds of higher molar mass have to be separated using HPLC

Gas Chromatography

- **Environmental monitoring** of **pollutants** in air, water and soil
- Detecting **dangerous gases** (e.g. CH_4, CO) in underground mines
- **Forensic** analysis
- Testing for **illegal/banned substances** (e.g. Drugs at sporting events)
- Analysis of petroleum/other **refined products**

HPLC

- *Requires that the **sample is soluble in a suitable solvent** and their **polarity allows separation***
- **Quality control** and analysis of fats, oils, vitamins, sugars, caffeine, antioxidants and medicine
- Analysis of amino acids, peptides and proteins in **biochemical research**

Pearson Ch 13

Saturday, 9 May 2020 2:21 PM

19.1 - Chromatography : An Introduction

- Instrumental chromatographs can analyse microlitre (10^{-6}L) amounts of a substance
- The can detect picogram (10^{-12}g) amounts of a substance in the samples
- Separation of components depends on solubility (which depend on intermolecular forces) in the mobile phase compared to their tendency to adsorb (stick to the surface of) to the stationary phase

19.2 - Paper Chromatography

- Stationary Phase: Water molecules bound to chromatography paper
- Mobile Phase: Liquid solvent
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- More polar components will adsorb to the stationary component as it is also very polar, while those with a greater tendency to remain dissolved in the mobile phase will move further
- If the components are colourless they can still be identified by using UV light or a locating agent (ninhydrin is very common)

19.3 - Column Chromatography

- Stationary Phase: Small beads of silica gel (SiO_2) or alumina (Al_2O_3)
- Mobile Phase: Liquid solvent
- Glass column is packed with the stationary phase and filled up with the mobile phase
- A small amount of the unknown substance is placed on top
- The stopcock is opened and the mobile phase will start dripping out of the column
- More solvent is added at the same rate it is flowing out
- The components will separate based on their polarity (more polar will adsorb to the stationary phase and move slower, less polar will dissolve in solvent better and move faster)

19.4 - Thin Layer Chromatography

- Stationary Phase: Glass plate coated with silica (SiO_2) or alumina (Al_2O_3)
- Mobile Phase: Liquid solvent
- Mobile phase moves up stationary phase through capillary action
- Mixture will separate based on polarity (more polar will adsorb to the stationary phase and less polar will dissolve better in the mobile phase)
- Produces better separation than paper chromatography
- Retention Factor (R_f) is used to identify substances
    - $R_f = \frac{distance\ moved\ by\ solute}{distance\ moved\ by\ solvent}$
- The R_f of a substance is dependent on the stationary and mobile phase as well as the temperature
    - To determine the substances, a TLC of the pure substances must be conducted under identical conditions
- Simple and inexpensive; used for separating inks, dyes, plant extracts and drug testing
    - If positive for drug testing, a confirmatory test is needed as it sometimes gives false positives

19.5 - Instrumental Chromatography

- Chromatography techniques in research and industry usually use techniques such as Gas Chromatography (GC) and High Performance Liquid Chromatography (HPLC)
- The principles of separation for these are the same as the simpler methods
    - Separation of the analytes (components of the mixture) depends on their attraction to the stationary vs mobile phase

19.6 - Gas Chromatography

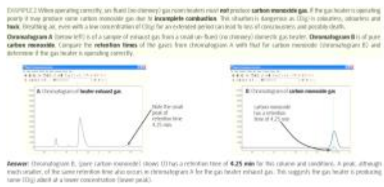
- **Stationary Phase:** High-boiling-point/non-volatile viscous liquid adsorbed onto solid particles such as silica
- **Mobile Phase:** Inert carrier gas
- Ideal for identifying individual substances and concentrations in mixtures
- *Not suitable for compounds that decompose when heated or don't vaporise easily (molecular*

mass above 300g per mol)

How it works

- The stationary phase is packed into a **stainless steel column** (usually 2-5mm diameter and 1.5-10m length)
  - Capillary columns are less than a millimetre in diameter and up to 100m length
- The liquid sample is **injected** into an injecting chamber via a microsyringe
  - The chamber is up to 250 degrees, and vapourises the sample which then flows with the MP into the column
- The column is in an **oven** and is set at a temperature to produce good separation of the components
- The different components will separate based on **volatility**; More volatile will have a greater tendency to stay in the MP and have a lower retention time, and vice versa
- As they reach the end of the column they are analysed by a detector which records them on a chromatogram
- **Retention time of a substance is the amount of time passed from injecting the sample to the detector recording a it passing out the end**
- It depends on **BP/volatility** as well as **column temperature**
  - A 10 degree increase in column temperature will approximately halve the retention time
  - The column temperature sometimes slowly increases so that the less volatile substances won't take so long to exit the column
- The **Flame Ionisation Detector (FID)** is a common detector used to detect substances exiting the column - gives retention time but doesn't identify the substance
- A Mass Spectrometer (MS) can be used as the detector to identify the substances
  - Known as **Gas Chromatography Mass Spectrometer (GCMS)**

19.7 - Identifying Analytes from Retention Time



- Peaks are characterised by **height**, **area** and **retention time**
- These features can be used to **identify analytes** and **determine their concentration**

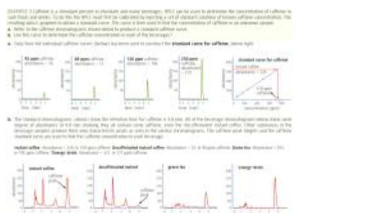
How it works

- A chromatogram of the sample of the pure substance being tested for must be produced under the **exact same conditions** (temp, gas and gas flow rate, column)
- The retention time of the pure substance is **compared** to the retention times of the unknown sample being tested

19.8 - High Performance Liquid Chromatography (HPLC)

- **Stationary Phase:** Fine solid particles (e.g. silica) packed into a column
- **Mobile Phase:** Liquid solvent
- If gas chromatography cannot be used for a particular sample, HPLC is usually used instead
- The column is usually 3-5mm diameter and 10-30cm length
- Particle size is very important as it presents a **larger surface area** for the analytes to **interact** with
- Separation based on **polarity**
  - In **Normal Phase HPLC** the SP is polar and MP is non-polar, so more polar analytes will have a longer retention time
  - In **Reverse Phase HPLC** (more common) the SP is non-polar and MP is polar, so more polar analytes will have a shorter retention time
  - Relative polarity of the SP and MP can be altered by slowly changing their composition over time so that analytes which aren't moving much will be released and flow out with the MP
- An **Ultraviolet Light Detector** is often used to detect the presence of analytes in the **eluent** (solution exiting the column)
  - Requires that the analytes absorb UV light

19.9 - Concentration Analysis from HPLC Chromatogram



- To determine the concentration of a substance, the chromatograph must be calibrated to it first
  - A **set of chromatographs** for **known concentrations** of the substance are created, and a graph of **absorbance vs concentration** is obtained from the data
- The **same chromatograph and conditions** are used on the unknown sample, and the absorbance of the corresponding retention time can be **plotted on the graph** obtained earlier to calculate the concentration

19.10 - Applications of GC and HPLC

- Both GC and HPLC are ideal for determining identity and concentration of substances within an unknown sample

- **Reliable**, **efficient**, relatively **fast** and **non-destructive (components can be collected)**
- Requires only minute samples

- GC requires the sample to be in a **vaporised state** so can only be applied to substances which **vapourise before decomposing** (generally molar **mass < 300g per mol**)
- Compounds of higher molar mass have to be separated using HPLC

Gas Chromatography

- **Environmental monitoring** of **pollutants** in air, water and soil
- Detecting **dangerous gases** (e.g. CH_4, CO) in underground mines
- **Forensic** analysis
- Testing for **illegal/banned substances** (e.g. Drugs at sporting events)
- Analysis of petroleum/other **refined products**

HPLC

- *Requires that the **sample is soluble in a suitable solvent** and their **polarity allows separation***
- **Quality control** and analysis of fats, oils, vitamins, sugars, caffeine, antioxidants and medicine
- Analysis of amino acids, peptides and proteins in **biochemical research**

Handouts

Wednesday, 6 May 2020 3:26 PM

Chromatography (Questions on front page)

1. What is the aim of chromatography?
   To separate a sample into its components to produce a pure fraction or quantify its components
2. What is the relation between retention time and the speed of the analysis?
   Longer retention time = slower speed of analysis. The retention time is a measure of how long a substance is held in a column, therefore the longer the time is the slower the speed of the chromatography is
3. How do we measure efficiency?
   Theoretical plate number - how good a column is at separating compounds
4. What are the two main phases in chromatographic techniques?
   Stationary and Mobile
5. What is the most important part of HPLC?
   The column packed with tiny particles of the stationary phase
6. In which phase do we use silica-based particles?
   Stationary
7. What is the most common HPLC technique?
   Ultraviolet absorption

Solubility

Monday, 11 May 2020 2:22 PM

Water as a Resource (Extension)

- Only 0.5% of water on Earth is drinkable and accessible - rest of freshwater is trapped in glaciers/soil
- Contamination of water supplies comes from farms, industry, mines, and soil
- Chemical contaminant: Elements/Compounds that may be harmful if consumed
  - Includes heavy metals, pollutants (fertiliser), organic pollutants

Removing Saline Solutes

Distillation

- Water is volatile, salts non-volatile
- Water evaporated from seawater and condensed
- Heat released from condensation used to heat incoming water
- Very energy demanding as it requires both vaporisation and condensation

Reverse Osmosis

- High pressure pump forces water through a semi-permeable membrane
- Ions, larger molecules, organisms and viruses can't get through the SPM
- Product called permeate
- Much more efficient than distillation as it doesn't require any state changes

Heavy Metals

- Severe health effects such as cancer, organ/nervous system damage, death
  - E.g. Copper from copper pipes may result in anaemia (lack of blood), liver/kidney damage and intestinal irritation
- Removed by precipitation - find soluble compound which makes insoluble precipitate with metal to be removed

Example : Tasmania (Mining)

- 2013, 1/3 of Tasmania's town water supplies didn't meet national regulations for heavy metals
- Residents of 5 towns were told not to drink the water due to contamination
- Contamination (zinc, cadmium and lead) came from nearby mining operations

Purification

- **Aeration:** Water sprayed in the air to reduce presence of gases (e.g. CO_2, H_2S)
- **Flocculation:** Small suspended particles made to join together to make larger particles which sink and settle to the bottom
  - Alum ($Al_2(SO_4)_3$) and lime ($Ca(OH)_2$) dissolved which form $Al(OH)_{3(s)}$, a gelatinous substance called floc
  - Fine particles and microorganisms are adsorbed in the precipitate, forming larger, heavier particles
- **Settling:** Water left to stand, floc settles at the bottom, removed
- **Filtration:** Water filtered through sand/gravel bed, removes remaining suspended matter
- **Chlorination:** Gaseous chlorine added to remove biological contaminants
- **Fluoridation:** Fluorine added to reduce tooth decay (reacts with enamel)
- **pH:** Addition of lime (to increase pH) or CO_2 (to decrease pH) used to achieve pH range of 6.5-8.5

Anomalous Properties of Water

- Each water molecule can have up to 4 hydrogen bonds (2 lone pairs on O + 2 H atoms)

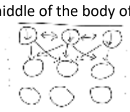
Relatively High Melting/Boiling Points

- Significantly more heat energy required to break hydrogen bonds in water than dipole-dipole/dispersion forces in other group 16 hydrides/similar sized molecules

Density in Liquid and Solid States

- As a liquid, each molecule generally has 1-2 hydrogen bonds at a time
- When cooled, molecules arrange themselves to each have their maximum 4 hydrogen bonds, making a much more open structure

High Surface Tension

- Surface Tension: A liquid's resistance to increasing its surface area
- Water molecules at the surface have an imbalance of forces, pulling them in towards the middle of the body of water
  - 

High Heat Capacity

- Heat Capacity: Ability of a substance to absorb and store heat energy
  - Higher heat capacity = more heat energy needed to increase its temperature
- High heat capacity due to hydrogen bonds, as they are stronger so can absorb more energy before they break

Latent Heat

- Latent Heat: Energy absorbed by a fixed amount of a substance as it changes from a solid to liquid/liquid to gas
  - I.e. The amount of energy required for an amount (usually 1 mole) of a substance to change state
- While it is changing state, temperature stays constant
- Latent heat of fusion: Solid to liquid
- Latent heat of vaporisation: Liquid to gas

Solutions

- **Solution:** A homogeneous mixture in which the solute particles are evenly distributed throughout particles of solvent
  - May be solid (e.g. Bronze), liquid (e.g. Sea water) or gas (e.g. Air)
- **Aqueous Solution:** Liquid solution where the solvent is water
- **Dissolving:** Solid particles are separated and distributed throughout the mixture
- **Crystallisation:** Dissolved particles return to solid form
- **Solubility:** The maximum amount of solute that can be dissolved in a given quantity of solvent (usually 100g water) at a certain temperature (usually 25°C)

Saturation

- **Unsaturated:** Contains less solute than the solvent is normally able to dissolve
- **Saturated:** Contains the same amount of solute that the solvent is normally able to dissolve at a given temperature
- **Supersaturated:** Unstable solution which contains more solute than the solvent is normally able to dissolve. If disturbed, solute will crystallise leaving a saturated solution

Trends

- Solubility of most solids increases with temperature (higher kinetic energy makes the

bonds between them easier to break)
- Solubility of Gases:
  - Decreases with temperature (higher kinetic energy causes molecules to break intermolecular forces and escape the solution)
  - Increases with pressure
  - Increase with polarity

Solvation

- **Solvation:** The process by which solvent molecules dissolve and surround solute particles
  - Occurs when the solvent-solute attraction is strong enough to overcome the solute-solute and solvent-solvent attraction
- **Hydration:** Solvation by water (i.e. Solvent is water)
- **Dissociation:** When an ionic compound dissolves, ions are separated
- **Ionisation**: When a molecular electrolyte dissolves, it separates and forms ions

Soluble or Insoluble?

- Energy is required to break solute and solvent bonds (endothermic)
- Energy is released as bonds form between solute and solvent (exothermic)
- If the process is favourable (doesn't require too much energy overall) the solution will form
- I.e. The solute-solvent bonds must be strong enough to overcome the solvent-solvent and solute-solute bonds

Electrolytes

- When dissolved, solutes can be present as mostly ions, mostly molecules, or a mix of both
- Classified as
  - **Strong Electrolytes:** Ionic compounds and strong acids
    - Single arrow shows ionisation/dissociation that goes to **completion**
    - $NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$
    - $H_2SO_4(aq) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$
  - **Weak Electrolytes:** Weak acids and weak bases
    - Double arrow shows **partial reaction**
    - $CH_3COOH(aq) \rightleftarrows H^+(aq) + CH_3COO^-(aq)$
  - **Non-Electrolytes:** None of the above

Precipitation Reactions

- **Precipitation reaction**: Forms a precipitate from a previously clear solution
- **Precipitate:** Insoluble solid formed during a precipitation reaction
- Precipitate forms as a suspension (makes it look cloudy)

Uses of Precipitation Reactions

- Identifying unlabelled substances (must have different solubility patterns)
  - E.g. Unlabelled $Pb(NO_3)_2$ and $Ca(NO_3)_2$
  - React one with Sodium Iodide ($NaI_{(aq)}$). PbI_2 is a yellow precipitate (substance is $Pb(NO_3)_2$)
- Removal of toxic chemicals in water treatment
- Coloured pigments for paints/dyes

Concentration

- Measure of the relative amount of solute dissolved in the solvent

Units of Concentration

- Moles per litre (mol L^{-1}); aka molarity/molar concentration
  - $concentration = \frac{n\ (solute)}{V\ (solution)}$
- Grams per litre (g L^{-1})
  - $concentration = \frac{m\ (solute)}{V\ (solution)}$
- Parts per million (ppm) used for solutes with very low concentration
  - $concentration = \frac{m\ (solute) * 1\,000\,000}{m\ (solution)}$
  - $concentration = \frac{m\ (solute)(mg)}{m\ (solution)\ (kg)}$
  - E.g. 3mol NaCl dissolved in 10L water
    - 0.3mol NaCl weighs 17.55g = 17 550mg
    - 10L water weighs 10 000g = 10kg
    - $c = \frac{175\,500}{10} = 1755ppm$
- %Volume
  - $concentration = \frac{V\ (solute)}{V\ (solution)} * 100\%$
- %Mass
  - $concentration = \frac{m\ (solute)}{m\ (solution)} * 100\%$

Dilution

- **Dilution:** Adding more solvent to a solution to reduce concentration
- $c_1V_1 = c_2V_2$: $concentration * volume = concentration\ (after\ dilution) * volume\ (after\ dilution)$
  - Moles of solute doesn't change, $n = cV$

Calculating Volumes for Dilution

- E.g. How much water needs to be added to an 18.4mol L^{-1} H_2SO_4 solution to prepare 5L of 1.5mol L^{-1} H_2SO_4 solution?
  - Use formula $c_1V_1 = c_2V_2$
    - $c_1 = 18.4mol\ L^{-1}$
    - $c_2 = 1.5mol\ L^{-1}; V_2 = 5L$
  - $V_1 = \frac{c2*V2}{c1} = \frac{7.5}{18.4} = 0.408L$
  - $Amount\ of\ solute = 0.408L$
  - $Amount\ of\ water = 5 - 0.408 = \mathbf{4.592L}$

All Notes Summary

Sunday, 31 May 2020 10:58 AM

Water as a Resource

- Only 0.5% of water on Earth is drinkable and accessible - rest of freshwater is trapped in glaciers/soil
- Reservoirs (fed by rivers) are the main source of household water in Australia
- Contamination of water supplies comes from farms, industry, mines, and soil
- Chemical contaminants include heavy metals, pollutants (fertiliser), organic pollutants

Purification

- **Aeration:** Water sprayed in the air to reduce presence of gases (e.g. CO_2, H_2S)
- **Flocculation:** Small suspended particles made to join together to make larger particles which sink and settle to the bottom
  - Alum ($Al_2(SO_4)_3$) and lime ($Ca(OH)_2$) dissolved which form $Al(OH)_{3(s)}$, a gelatinous substance called floc
  - Fine particles and microorganisms are adsorbed in the precipitate, forming larger, heavier particles
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- **Filtration:** Water filtered through sand/gravel bed, removes remaining suspended matter
- **Chlorination:** Gaseous chlorine added to remove biological contaminants
- **Fluoridation:** Fluorine added to reduce tooth decay (reacts with enamel)
- **pH:** Addition of lime (to increase pH) or CO_2 (to decrease pH) used to achieve pH range of 6.5-8.5

Anomalous Properties of Water

- Each water molecule can have up to 4 hydrogen bonds (2 lone pairs on O + 2 H atoms)

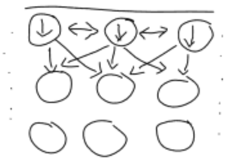
Relatively High Melting/Boiling Points

- Significantly more heat energy required to break hydrogen bonds in water than dipole-dipole/dispersion forces in other group 16 hydrides/similar sized molecules

Density in Liquid and Solid States

- As a liquid, each molecule generally has 1-2 hydrogen bonds at a time
- When cooled, molecules arrange themselves to each have their maximum 4 hydrogen bonds, making a much more open structure

High Surface Tension

- Surface Tension: A liquid's resistance to increasing its surface area
- Water molecules at the surface have an imbalance of forces, pulling them in towards the middle of the body of water
  - 

High Heat Capacity

- Heat Capacity: Ability of a substance to absorb and store heat energy
  - Higher heat capacity = more heat energy needed to increase its temperature
- High heat capacity due to hydrogen bonds, as they are stronger so can absorb more energy before they break

Latent Heat

- Latent Heat: Energy absorbed by a fixed amount of a substance as it changes from a solid to liquid/liquid to gas
  - I.e. The amount of energy required for an amount (usually 1 mole) of a substance

    to change state
    - While it is changing state, temperature will stay constant
    - Latent heat of fusion: Solid to liquid
    - Latent heat of vaporisation: Liquid to gas

Solutions

- **Solution:** A homogeneous mixture in which the solute particles are evenly distributed throughout particles of solvent
    - Can be solid (e.g. Bronze), liquid (e.g. Sea water) or gas (e.g. Air)
- **Aqueous Solution:** Liquid solution where the solvent is water
- **Solvent:** The major component of a solution by mass
- Solute: The minor component of a solution by mass
- **Dissolving/Dissolution:** Solid particles are separated and distributed throughout the mixture, increasing solution concentration
- **Crystallisation:** Dissolved particles return to solid form, reducing solution concentration
- **Solubility:** The maximum amount of solute that can be dissolved in a given quantity of solvent (usually 100g water) at a certain temperature

Saturation

- **Unsaturated:** Contains less solute than the solvent is normally able to dissolve
- **Saturated:** Contains the same amount of solute that the solvent is normally able to dissolve
    - Rate of dissolving and crystallisation are equal
- **Supersaturated:** Unstable solution which contains more solute than the solvent is normally able to dissolve. If disturbed, solute will crystallise leaving a saturated solution
    - Can be achieved through heating and dissolving, then carefully cooling the solution

Trends

- Solubility of Solids:
    - *Generally* increases with temperature (higher kinetic energy makes the wbonds between them easier to break)
- Solubility of Gases:
    - Decreases with temperature (higher kinetic energy causes molecules to break intermolecular forces and escape the solution)
    - Increases with pressure
    - Increase with polarity

Solvation

- **Solvation:** The attraction of polar solvent molecules to both the positive and negative ions of an ionic solid
    - Solvation occurs when the solvent-solute attraction is strong enough to overcome the solute-solute and solvent-solvent attraction
- **Hydration:** Solvation by water, formed through ion-dipole forces of attraction
- **Dissociation:** When an ionic compound dissolves, ions are separated
- **Ionisation**: When a molecular electrolyte dissolves, it forms ions
    - Molecular compounds that can form hydrogen bonds are often soluble in water
- A solution will form if the solute-solvent attraction is strong enough to overcome the solvent-solvent and solute-solute attraction
- **Ion-Dipole Forces:** Electrostatic forces of attraction between ions and polar solvent molecules

Like Dissolves Like

- DON'T USE PHRASE IN ASSESSMENTS
- Polar solvents and polar solute can form dipole-dipole bonds (and sometimes hydrogen bonds) which favours dissolving)
    - Non-polar substances can only form dispersion forces which aren't strong enough to dissolve polar solutes or in polar solvents
- Non-polar solvents dissolve non-polar solutes as they all only form dispersion forces, no strong bonds required to be broken

Electrolytes

- When dissolved, solutes can be present as mostly ions, mostly molecules, or a mix of both
- Single arrow shows ionisation/dissociation that goes to **completion** (strong electrolytes)
- Double arrow shows **partial reaction** (weak electrolytes)
- Classified as
  - **Strong Electrolytes:** All/Mostly ions
    - Ionic compounds and strong acids
    - Ions dissociate
    - $NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$
    - $H_2SO_4(aq) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$
  - **Weak Electrolytes:** Partly ions partly molecules
    - Weak acids and weak bases
    - $CH_3COOH(aq) \rightleftarrows H^+(aq) + CH_3COO^-(aq)$
  - **Non-Electrolytes:** All/Mostly molecules
    - Not ionic, acid or weak base

Ionic Solubility

- High polarity of water makes it a good solvent for many ionic compounds
- High solubility of many ionic compounds in water is due to the many strong ion-dipole forces overcoming ionic forces between ions

Precipitation Reactions

- **Precipitation reaction**: A reaction that forms a precipitate from a previously clear solution
- **Precipitate:** An insoluble solid that forms during a precipitation reaction
- During the reaction the precipitate is in the form of a suspension
  - I.e. Lots of tiny particles spread out throughout the mixture (makes it look cloudy)
  - Over time suspended particles will settle to the bottom

Uses of Precipitation Reactions

- Identifying unlabelled substances
  - Only works for substances with different solubility patterns
- Removal of toxic chemicals in water treatment
- Coloured pigments for paints/dyes

Spectator Ions

- Don't take part in any chemical changes
- **Net Ionic Equation:** Chemical equation omitting spectator ions
- **ALWAYS USE NET IONIC EQUATION**

Concentration

- Concentration is a measure of the amount of solute dissolved in the solvent
  - Ratio of $\frac{quantity\ of\ solute}{quantity\ of\ solution}$

Units of Concentration

- Moles per litre (mol L^{-1}); aka molarity/molar concentration
  - $concentration = \frac{Moles\ (n)}{Volume\ of\ Solution\ (V)}$
- Grams per litre (g L^{-1})
  - $concentration = \frac{mass\ of\ solute\ (m)}{volume\ of\ solution\ (V)}$
- Parts per million (ppm) used for solutes with very low concentration
  - $concentration = \frac{mass\ of\ solute\ * 1\,000\,000}{mass\ of\ solution}$
  - E.g. 3mol NaCl dissolved in 10L water
    - 3mol NaCl weighs 175.5g = 175 500mg
    - 10L water weighs 10 000g = 10kg

- ☐ $c = \frac{175\,500}{10} = 17550ppm$
- %Volume
  - $concentration = \frac{V\ (solute)}{V\ (solvent)} * 100\%$
- %Mass
  - $concentration = \frac{m\ (solute)}{m(solvent)} * 100\%$

Dilution

- **Dilution:** Process of adding more solvent to a solution
- $c_1V_1 = c_2V_2$: $concentration * volume = concentration\ (after\ dilution) * volume\ (after\ dilution)$
  - Moles of solute doesn't change, $n = cV$

Calculating Volume when Solution Diluted

- E.g. How much water needs to be added to an 18.4mol L^{-1} H_2SO_4 solution to prepare 5L of 1.5mol L^{-1} H_2SO_4 solution?
  - Use formula $c_1V_1 = c_2V_2$
    - ☐ $c_1 = 18.4mol\ L^{-1}$
    - ☐ $c_2 = 1.5mol\ L^{-1}; V_2 = 5L$
  - $V_1 = \frac{c1*V1}{c2} = \frac{7.5}{18.4} = 0.408L$
  - $Amount\ of\ solute\ = 0.408L$
  - $Amount\ of\ water\ = 5 - 0.408 = \mathbf{4.592L}$

Class

Monday, 11 May 2020 2:22 PM

Solutions

Precipitation

Concentration

Hydrated Compounds

- **Hydrated compound:** An ionic compound that contains water within its crystal structure
  - Water can be driven off by heating the compound, resulting in an **anhydrous** substance
- **Water of Crystallisation:** Water molecules that make up part of the structure of an ionic crystal
- E.g. Copper(II) Sulfate Pentahydrate - $CuSO_4.5H_2O$

Lucarelli Ch 16 - Solutions

Monday, 11 May 2020 2:22 PM

16.1 - Solutions and the Dissolving Process

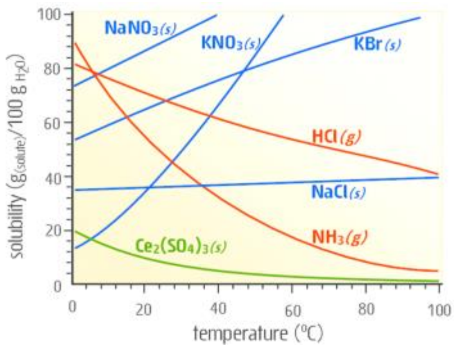
- **Solution:** A mixture where the solute particles are homogeneously distributed through particles of solvent
    - **Solvent:** The major component by mass
- Solutions can be solid (e.g. Bronze), liquid (e.g. Sea water) or gas (e.g. Air)
- **Aqueous Solutions:** Liquid solutions where the solvent is water
- **Dissolving:** Solid particles are separated and distributed throughout the mixture, increasing solution concentration
- **Crystallisation:** Returns dissolved particles to solid form, reducing solution concentration

Unsaturated, Saturated and Supersaturated

- **Unsaturated:** Contains less solute than the solvent is normally able to dissolve
- **Saturated:** Contains the same amount of solute that the solvent is normally able to dissolve
    - When the speed of dissolving and crystallisation are equal
- **Supersaturated:** Contains more solute than the solvent is normally able to dissolve

Solubility and Heat Trends

- Solubility is given in $g_{(solute)}\ 100mL^{-1}{}_{(H2O)}$ (i.e. Grams of solute dissolved per 100mL water)
- Most solids ($Ce_2(SO_4)_3$ exception) increase in solubility with temperature
- Gases are always less soluble at higher temperatures



16.2 - Solutes and Ion Formation

- When dissolved, some solutes are present as only/mostly ions, some only/mostly molecules, and some partly molecules partly ions
- These are classified as:
    - **Strong Electrolytes:** Only/Mostly ions
    - **Weak Electrolytes:** Ions and molecules
    - **Non-Electrolytes:** Only/Mostly molecules

Determining the Type of Electrolyte

1. If it is an ionic compound (e.g. Sodium Chloride) or a strong acid (e.g. Hydrochloric Acid, Nitric Acid) it is a **strong electrolyte**
    - $NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$
    - $H_2SO_4(aq) \rightarrow 2H^+(aq) + SO_4^{2-}(aq)$
2. If it is a weak acid (e.g. Phosphoric Acid) or a weak base (e.g. Ammonia) it is a **weak electrolyte**
    - Double arrow is used to show weak electrolytes
    - $CH_3COOH(aq) \rightleftarrows H^+(aq) + CH_3COO^-(aq)$
3. If none of the above, it is a **non-electrolyte**
    - $C_6H_{12}O_6(s) \rightarrow C_6H_{12}O_6(aq)$

16.3 - Equations of Solvation

- When an ionic compound dissolves, the ions are released into the water in a process called **dissociation**
- When a molecular electrolyte dissolves, it forms ions in a process called **ionisation**
- **Single arrow** shows an ionisation or dissociation reaction that goes to completion
    - Strong electrolytes
- **Double arrow** shows an ionisation reaction that does not go to completion (compound continuously forming the molecule and breaking apart)
    - Weak electrolytes
- Pure water is not a conductor of electricity, but when it contains a dissolved electrolyte the ions from the electrolyte will be free to move, allowing it to conduct electricity
    - **Electrical conductivity** of a solution depends on the ability of cations to move towards a negative electrode and anions to move toward a positive electrode
    - Stronger electrolytes will produce a stronger conductive solution than weaker electrolytes of the same concentration

16.4 - Ionic Solubility and Solvation

- Solubility of ionic compounds is largely dependent on the solvation properties of the solvent
- **Solvation:** The process by which solvent molecules surround solute molecules
    - In a polar solvent it involves the attraction of polar solvent molecules to the positive and negative ions of ionic solids
- The high polarity of water allows it to be a good solvent for many ionic compounds
- **Hydration:** Solvation by water, which occurs because of electrostatic ion-dipole forces of attraction between a dipole of water and an ion
    - Slightly positive H atom attracted to negative ion
    - Slightly negative O atom attracted to positive ion
    - Each ion is surrounded by multiple water molecules
- High solubility of many ionic compounds in water is due to the strong ion-dipole forces overcoming the electrostatic forces between the oppositely charged ions

16.5 - Precipitation Reactions

- A **precipitation reaction** is when an insoluble solid forms within a previously clear solution
- One way this can happen is if two different ionic solutions are mixed
    - The free ions can form bonds with each other, and if any of the possible substances are insoluble they will form a precipitate
- During the precipitation reaction the solute is in the form of a suspension

- Lots of tiny particles of solid which are spread out through the mixture (makes it look cloudy)
- Over time the suspended particles will settle to the bottom of the solution

16.6 - Using Precipitation to Distinguish Compounds

- Reacting an unknown solution with a known solution can determine whether a particular ion is present or not
- Can be used if e.g. Two solutions got mixed up and you need to find out which is which (e.g. $Pb(NO_3)_2$ and $Ca(NO_3)_2$)
- E.g. To check if there are Pb^{2+} ions, react it with Sodium Iodide ($NaI_{(aq)}$). If Pb^{2+} ions are present a yellow precipitate will form. This tells you it is $Pb(NO_3)_2$ and not $Ca(NO_3)_2$
- Only works if the substances being tested for have ions with different solubility patterns

16.7 - Solution Concentration

- Usually a ratio of $\frac{quantity\ of\ solute}{quantity\ of\ solution}$
- If using units, they vary depending on the situation
  - Moles per litre (mol L^{-1}) are often used in laboratories
    - $concentration\ = \frac{Moles\ (n)}{Volume\ of\ Solution\ (V)}$
    - c: Concentration of solute in mol L^{-1}
    - V: Volume of solution (solute + solvent) in Litres
  - Percentage by mass per litre (g L^{-1}) used for packaging of consumer goods
    - $concentration = \frac{mass\ of\ solute\ (m)}{volume\ of\ solution\ (V)}$
    - m: mass of solute in grams
  - Parts per million (ppm) used for solutes with very low concentration
    - $concentration = \frac{mass\ of\ solute\ * 1\ 000\ 000}{mass\ of\ solution}$

16.8 - Ion Concentration in Strong Electrolyte Solutions

- Concentration of an ion from an ionic compound in a solvent is the concentration of the ionic compound multiplied by the ion's subscript of its chemical formula

EXAMPLE 7 What is the concentration of all of the ions in the following two solutions?

a. 3.1 mol L^{-1} $Al_2(SO_4)_3(aq)$
b. 0.80 mol L^{-1} $(NH_4)_2Fe(SO_4)_2(aq)$

a. **$c[Al^{3+}(aq)]$** = 2 x $c[Al_2(SO_4)_3]$ = 2 x 3.1 = **6.2 mol L^{-1}** As Al^{3+} has a subscript of 2
$c[SO_4^{2-}(aq)]$ = 3 x $c[Al_2(SO_4)_3]$ = 3 x 3.1 = **9.3 mol L^{-1}** and SO_4^{2-} has a subscript of 3.

b. **$c[NH_4^+(aq)]$** = 2 x $c[(NH_4)_2Fe(SO_4)_2]$ = 2 x 0.80 = **1.6 mol L^{-1}** Since NH_4^+ has a subscript of 2,
$c[Fe^{2+}(aq)]$ = 1 x $c[(NH_4)_2Fe(SO_4)_2]$ = 1 x 0.80 = **0.80 mol L^{-1}** Fe^{2+} has a subscript of 1 and
$c[SO_4^{2-}(aq)]$ = 2 x $c[(NH_4)_2Fe(SO_4)_2]$ = 2 x 0.80 = **1.6 mol L^{-1}** SO_4^{2-} has a subscript of 2.

16.9 - Need a Drink?

- Of all the water on Earth,
  - 97.3% is saline,
  - 2.7% is fresh water,
  - 0.6% is available as freshwater in rivers, lakes, swamps or groundwater as a large amount of freshwater is in polar ice-caps and glaciers
- In our bodies, water is a solvent for metabolic processes, transporting nutrients, excretion of cellular waste and regulating body temperature
- Water that is safe to drink is called potable water
  - Clear, colourless, odourless, contains no pathogens/toxic substances, pleasant to taste
- Most of our water now comes from desalination plants (water from the ocean), groundwater and surface water

16.10 - Potable Water from Seawater

- Seawater has a salinity of over 35 000mg L^{-1} total dissolved solids (TDS), so it is unusable for drinking, agriculture, washing, etc.
- Distillation and Reverse Osmosis (RO) are two methods of desalinating seawater

Distillation

  - Water is volatile, salts non-volatile
  - Water is evaporated from the seawater and condensed
  - Heat released from the condensation of water vapour is used to heat the incoming salt water
  - Even recycling heat, this is a very energy demanding process

Reverse Osmosis (RO)

  - Uses high pressure pumps to force water through a semi-permeable membrane (SPM)
  - Ions, larger molecules, cellular organisms and viruses can't pass through the SPM
  - Saltwater going in is called the *feed water*, product called *permeate*
  - The permeate is distilled water, so CO_2 and CaO are added for taste, health reasons, and to give it a suitable pH
  - Requires about 1/3 of the energy used in the most efficient distillation plants

16.11 - Groundwater Treatment

- Common processes for treating groundwater:
  - **Aeration:** Spraying the water in the air to reduce the presence of gases (e.g. CO_2, H_2S)
    - Extra oxygen along with added chlorine remove dissolved organic compounds
  - **Clarification:** After aeration; removes suspended particles by adding lime (CaO)
  - **Sand Filtration:** Water passed through beds of coarse sand which adsorbs organic compounds that would otherwise colour water

- **Disinfection:** Chlorine and ammonia used together to destroy harmful bacteria/viruses
- **Fluoridation:** Addition of fluorine to water as it is good for the enamel on teeth, reducing cavity rates
- **pH:** Addition of lime (to increase pH) or CO_2 (to decrease pH) used to achieve pH range of 6.5-8.5

Pearson Ch 15 - Water

Monday, 11 May 2020 2:22 PM

15.1 - Essential Water

- Only 2.5% of water on Earth is drinkable, only 0.5% of water on Earth is drinkable and accessible
- Reservoirs (fed by rivers) are the main source of household water in Australia
- Contamination of water supplies comes from farms, industry, mines, and soil
  Chemical contaminants include heavy metals, pollutants (fertiliser), organic pollutants

Purification

- Flocculation: Small suspended particles made to join together to make larger particles which sink and settle to the bottom
  - Achieved by adding alum ($Al_2(SO_4)_3$) and lime ($Ca(OH)_2$) which react to precipitate $Al(OH)_{3(s)}$, a gelatinous substance called floc
  - Fine particles and microorganisms are adsorbed in the precipitate, forming larger, heavier particles
- Settling: Water left to stand to allow floc to settle at the bottom, removed
- Filtration: Water filtered through a bed of sand and gravel, removing any remaining suspended matter
- Chlorination: Water treated with gaseous chlorine to kill bacteria/remove biological contaminants
- Fluoridation: Fluorine added to reduce tooth decay
  - Added in the form of compounds including NaF, Na_2SiF_6, H_2SiF_6 which dissolve in water
  - Reacts with enamel ($Ca_{10}(PO_4)_6(OH)_2$), replacing the hydroxyl ion, forming fluorapatite ($Ca_{10}(PO_4)_6F_2$) which is stronger and more decay resistant

15.2 - Properties of Water

Water and Hydrogen Bonding

Structure of Water

- Each hydrogen atom single covalent bond to the oxygen atom
- Oxygen atom 2 lone pairs of electrons
- Bent shape (104.5 degrees)
- Polar bonds (Oxygen slightly negative; Hydrogen slightly positive)
- Overall is a dipole molecule

Hydrogen Bonding

- Main type of intermolecular force is hydrogen bonds
- Each water molecule can have up to 4 hydrogen bonds (2 lone pairs on O + 2 H atoms)

Relatively High Melting/Boiling Points

- Excluding water, group 16 hydrides increase in melting/boiling points down the group
  - Due to larger size (stronger dispersion forces)
- Significantly more heat energy is required to break the hydrogen bonds in water

Density in Liquid and Solid States

- As a liquid, each water molecule tends to only make 1-2 hydrogen bonds at a time
- When it is cooled, the molecules slow down, and eventually arrange themselves to each have 4 hydrogen bonds, leaving a much more open structure

High Surface Tension

- Surface Tension: The resistance of a liquid to increasing its surface area
- Water molecules at the surface aren't completely surrounded by other water molecules
- Forces from the hydrogen bonds pull it towards the centre of the body of water
  - On average each molecule only spends one nanosecond (10^{-9} seconds) at

the surface before being pulled down

Heat Capacity

- Heat Capacity: Ability of a substance to absorb and store heat energy
  - I.e. How much energy a substance can absorb as its temperature increases
- If the same amount of heat energy is applied to substances with different heat capacities, their temperatures will change by different amounts
- Water has a relatively high heat capacity so it takes more energy to heat water up than many other substances

Specific Heat Capacity

  - Amount of energy required to increase the temperature of a certain amount (usually 1g) by 1 degree Celsius (Joules per gram per degrees Celsius - $Jg^{-1} C^{-1}$)
  - Reflects the type of bonding holding the molecules/ions/atoms together
    - For covalent molecules it depends on the strength of the intermolecular forces
  - Common Substances:

Substance	Specific Heat Capacity $Jg^{-1} C^{-1}$
Water	4.18
Ethanol	2.40
Sand	0.48
Copper	0.39
Lead	0.16

Specific Heat Capacity of Water

    - Water has a high specific heat capacity due to the presence of hydrogen bonds
    - Hydrogen bonds are stronger than other intermolecular forces, can absorb more energy before they break
    - 1mL of water = 1g of water

Latent Heat

- Latent Heat: The energy absorbed by a fixed amount of substance as it changes state from a solid to a liquid/liquid to a gas at its melting/boiling point
- I.e. It is the amount of energy required to change an amount of substance (usually 1 mole) from a solid to liquid or liquid to gas
- While it is changing state, its temperature will not change

Values of Latent Heat

  - Latent heat of fusion = heat needed to change 1 mole from solid to liquid
  - Latent heat of vaporisation = heat needed to change 1 mole from liquid to gas

Latent Heat of Water

  - Latent heat of fusion = 6.0kJ mol^{-1}
  - Latent heat of vaporisation = 44.0kJ mol^{-1}
  - High latent heat due to strong hydrogen bonding

15.3 - Water as a Solvent

- When particles dissolve in water they are free to move throughout the solution
- When two aqueous reactants are combined in a reaction vessel the particles are free to react with each other

Forces between Solute and Solvent Particles

- An **aqueous solution** is formed when a solid, liquid or gas is dissolved in water
- In all solutions:
  - Solute and solvent cannot be distinguished (they are **homogeneous**)
  - Dissolved particles are too small to see
  - Amount of dissolved solute varies from one solution to another

Process of Dissolving

- The process of dissolving a solute in a solvent is called **dissolution**
- If the solute and solvent are liquids they are **miscible**
- Processes that occur during dissolution:
  - Particles of solute separated from each other
  - Particles of solvent separated from each other
  - Solute and solvent particles attracted to each other
  - Solute particles surrounded by solvent particles and carried throughout the solution

Forces Involved in Dissolving

- Three important forces to consider if a substance will dissolve or not
  - Forces holding solute particles together
  - Forces holding solvent molecules together (e.g. hydrogen bonds in water)
  - Forces that would potentially form between solute and solvent particles if the substance dissolves
- For a solute to dissolve in a solvent (be soluble), the solute-solvent attraction must be stronger than the solute-solute and solvent-solvent attraction
- "Like dissolves like" - polar solvents will generally dissolve polar solutes but not non-polar solutes, and non-polar solvents will generally dissolve non-polar solutes but not polar solutes

Summary

- **Solution:** A homogeneous mixture in which the solute is evenly distributed throughout the solvent
- **Aqueous solution:** A solution in which the solvent is water
- A solution forms when the solvent-solute attraction is strong enough to compete with the solvent-solvent and solute-solute attraction
- "Like dissolves like" - polar solvents generally dissolve polar solutes, non-polar solvents generally dissolve non-polar solutes

15.4 - Water as a Solvent of Molecular Substances

Molecular Compounds that form Hydrogen Bonds with Water

- Molecular compounds that can form hydrogen bonds are often soluble in water
- E.g. Ethanol: $C_2H_5OH_{(l)} \xrightarrow{H2O} C_2H_5OH_{(aq)}$; All sugars
  - State symbol is the only thing that changes to show it is now dissolved
  - H_2O sits above the arrow since there isn't any direct reaction between water and the solute
- More polar molecules are more likely to dissolve in water
- Some molecules (e.g. Ethanol) have a polar and non-polar section
  - Larger non-polar section means it is less soluble in water
- Non-polar molecules only have dispersion forces with water, not strong enough to overcome hydrogen bonds between water molecules

Molecular Compounds that Ionise

- Very polar covalent bonds break when dissolved in water
- Strong acids will **ionise** when dissolved in water
- E.g. HCl
  1. Polar bonds between H and Cl are broken producing H^+ and Cl^-
  2. Covalent bond forms between each H^+ and H_2O molecule forming H_3O^+
  3. Ion-dipole attractions form between the H_3O^+ and Cl^- ions
     - $HCl_{(g)} + H_2O_{(l)} \rightarrow H_3O^+_{(aq)} + Cl^-_{(aq)}$

Summary

- Water is a very good solvent for polar compounds
- Some dissolve by forming hydrogen bonds
- Some dissolve by ionising

15.5 - Water as a Solvent of Ionic Compounds

- Ionic substances dissolve in water by dissociation
    - Water molecules form ion-dipole bonds with the positive and negative ions, pulling them away from their lattice
- Some substances are insoluble as the ion-dipole attractions aren't strong enough to overcome the ionic bonding
- Substances are on a scale of solubility, not just 'insoluble' or 'soluble'
- Substances containing Na^+, K^+, NH_4^+, NO_3^- or CH_3COO^- are always soluble

15.6 - Solubility

- **Solubility:** The maximum amount of solute that can be dissolved in a given quantity of solvent (usually 100g) at a certain temperature
- **Unsaturated:** more solute can be dissolved
- **Saturated:** No more solute can be dissolved
- **Supersaturated:** Unstable solution which contains more dissolved solute than a saturated solution. If disturbed, some solute will separate from the solvent and form a solid

Solubility Curves

- Mass of solute dissolved per 100g (100mL) of water; $g\ 100mL^{-1}$
- Solubility of many substances changes with temperature
- When increasing temperature:
    - Solids generally increase in solubility
    - Gases always decrease in solubility

Crystallisation

- Occurs when an unsaturated solution becomes saturated and crystals form
- Two main methods:
    1. Cooling a solution may reduce solubility to a point where it becomes supersaturated and crystals will form
    2. Evaporating the solvent

Variables Affecting Crystal Growth

- Rate of cooling: Slow cooling produces larger crystals, fast cooling produces smaller crystals
- Rate of evaporation (solvent): Slow evaporation produces larger crystals, fast evaporation produces smaller crystals
- Nucleation: If a 'nucleus' (solid) is present in the solution (can even just be a fleck of dust) the crystals will form around it much faster

Pearson Ch 16 - Aqueous Solutions

Tuesday, 12 May 2020 8:46 PM

16.1 - Precipitation Reactions

- A **precipitation reaction** occurs when two solutions are mixed and a solid product (precipitate) is formed
- **Spectator ions:** Ions that are not involved in the formation of the precipitate and don't undergo a chemical change
- **Net Ionic Equation:** Chemical equation leaving out spectator ions

16.2 - Concentration of Solutions

- The **concentration** of a solution is the **relative amount of solute to solvent** in a solution

Units of Concentration

- Units are always made up of two parts:
  - How much solute there is
  - How much solution there is
- g L^{-1} (grams per litre)
- mol L^{-1} (moles per litre)
- ppm (parts per million)
  - Mass of solute in grams per 1000 000 grams of solution
- Percentage by mass

16.3 - Molar Concentration

- Allows calculations of amount of moles of particles present in a solution
- Often called **molarity**
- Moles per litre (mol L^{-1})
- $c = \frac{n}{V}$ can be rearranged to $n = cV$ depending on the information given
  - $c_{(\text{mol L-1})}$: concentration (mol L^{-1})
  - n: moles (mol)
  - V: Volume (L)

16.4 - Dilution

- Process of adding more solvent to a solution is called **dilution**
- $c_1V_1 = c_2V_2$
  - $concentration * volume = concentration\ (after\ dilution) * volume\ (after\ dilution)$

Calculating Volume when Solution Diluted

- Used to determine how much solute/solvent is needed to dilute a concentrated solution to a given level
- E.g. How much water needs to be added to an 18.4mol L^{-1} H_2SO_4 solution to prepare 5L of 1.5mol L^{-1} H_2SO_4 solution?
  - Use formula $c_1V_1 = c_2V_2$
    - c_1 = 1.5mol L^{-1}; V_1 = 5L
    - c_2 = 18.4mol L^{-1}
  - $V_2 = \frac{c1 * V1}{c2} = \frac{7.5}{18.4} = 0.408L$
  - $Amount\ of\ solute\ = 0.408L$
  - $Amount\ of\ water\ = 5 - 0.408 = \mathbf{4.592L}$

16.5 - Calculations involving Reactions in Solutions

- If quantity of one reactant or product is known, the quantity of all other substances can be determined

1. Write balanced chemical equation
2. Calculate moles of given substance
3. Calculate moles of unknown using coefficients for the ratio
4. Use $n = \frac{m}{M}$ or $n = \frac{c}{V}$ to calculate the answer depending on the question (may need to rearrange formulae)

CAP 3 - Water Treatment

Thursday, 9 July 2020 11:17 AM

Water Treatment

Solubility Summary

CSIRO Booklet

Intro

- Poor water quality can lead to:
  - Pollutants such as metals and pathogens entering the food chain
  - Reduced agricultural productivity
  - Threatening human health and other life - ecosystems have adapted to their natural water quality; pollution decreases the water quality
- To manage pollutants:
  1. Define the uses and environmental values of water and the risks that pollution poses
  2. Identify the sources of pollution
     - Chemical/Biological pollutants may be transformed as they pass through the environment
  3. Set targets for improved water quality + actions to take to meet said targets
  4. Monitor water quality to identify new risks + evaluate effectiveness of actions
- Diffuse pollution from catchment water (rainfall?) is harder to solve and poses the most pollution problems today

Salinity

- Salinity: Amount of dissolved salt in the water
- Decreases usefulness of water for drinking, irrigation and farming
- Ultimate source is rainfall
- Clearing trees reduces evaporation, increasing the amount of ground salt being discharged to rivers
- Can be decreased by revegetation and making pastures with deep roots

Algal Blooms

- Eutrophication: excessive richness of nutrients/minerals in a body of water
  - Many rivers, lakes and coastal waters have become enriched in **nitrogen** and **phosphorous**
- Leads to overly rapid growth of algae (algal blooms)
- Large amount of blue-green algae excrete toxins hazardous to animals and people
- Rapid decomposition of the blooms consumes dissolved oxygen, killing fish
- Many pools and reservoirs are stratified (water separated into layers) under warm conditions
  - Bottom layer becomes oxygen deficient, causes the nitrogen and phosphorous to dissolve into the water - eutrophication (stimulates algal blooms)
  - To prevent this, many pools/reservoirs are mechanically stirred to reduce stratification

Sediments

- Turbidity: How transparent water is (less transparent = more turbid)
- Sediments have significant ecological effects
  - Can cover bed, covering habitats, plants, and other organisms
  - May resuspend, making water more turbid, blocking sunlight
  - Metals/Nutrients from sediment may be released into the water (eutrophication)
- Redox reactions may be used to clean up some sediments

Estuaries and Coastal Waters

- Estuary: The area where a river meets a stream
- Eutrophication and high turbidity reduces light levels, favouring algae over

seagrass and coral

Organic Chemicals and Pesticides

- Man-made chemicals may enter waterways as runoff, deposition from the air, or discharge of treated wastewaters from sewerage plants and industry
- Many of these chemicals (especially those used in the past) are very persistent and can still be found in water and sediments
- Reducing pollution from pesticides and other chemicals may be achieved in 3 ways:
  - Replacing them with less persistent chemicals
  - Recycling/storing water onsite to prevent discharge
  - Changing agricultural/industrial practices
- Genetically modified cotton reduces the use of pesticides as they contain proteins that give the plant pesticidal properties

Pathogens

- Pathogen: Disease causing organisms that end up in water due to discharge of sewerage effluents or as diffuse inputs from animal waste
- Very strict regulations ensure drinking water is free from pathogen contamination
  - E.g. Land around urban water catchments remain largely natural vegetation, so livestock waste doesn't contaminate the water

Metal Contaminants

- Metals are persistent (don't break down with time), so prevention at the source is preferable
- Contamination may result in increased metal concentration in organisms
  - Even in parts per billion concentrations may be toxic
- Mining sites provide low concentrations of metal, but must meet strict regulations
- Residential areas are diffuse sources trace concentrations of metals (ppb), which lead to a complex mixture of metal contaminants
  - E.g. Stormwater runoff from roads contains Zn from tyres & Cu from brake linings
- Free metal cations (e.g. Cu^{2+}) are particularly toxic

Acid Sulfate Soils

- When excavated or drained, soils containing sulfides react with water and oxygen in the air to form sulfuric acid
- Acid can dissolve metals (e.g. Al), and if discharged to rivers/estuaries, combination of acidity and metals can kill plants and animals, contaminate drinking water/food, and corrode concrete/steel
- Sulfide soils associated with coastal regions and particularly mangrove sediments

Groundwater Contamination

- Occurs through accidental spills/other unwanted releases of chemicals
- Move down through soil into groundwater below
- Pollutes drinking supplies, irrigation water, and ecosystems where groundwater discharges to other water bodies
- Slow movement + lack of dilution means high concentrations may be preserved over a long period of time (decades to centuries) - prevention at source preferable
- Pollutants include organic liquids (e.g. Petroleum/industrial solvents)
- Remediation:
  - Biodegradation: using bacteria that consumes organic contaminants
  - Artificial barrier: across leading edge of pollution
  - Permeable reactive barriers: allows water through, contains active ingredients that degrade/immobilise contaminants

Emerging Contaminants

- A wide range of chemicals in residential, industrial and agricultural wastewaters appear as mixtures at low concentrations in rivers and streams
  - Include: natural/synthetic hormones, insecticides, fire retardants, veterinary drugs
- Organic contaminants may disrupt animal reproduction/growth by interfering with hormones
  - Called "Endocrine Disrupting Chemicals"
  - Even if hormones are excreted from the body in a deactivated form,

chemical processes during treatment may restore them to their original form (e.g. Estrogen)
  - Nanomaterials are a new class of contaminants
    - Extremely varied composition, and different properties to their bulk material
    - Lots of research to determine whether they need their own quality guidelines

Course Outline (Solubility)

1. **Explain the unique properties of water using understanding of its molecular shape and ability to form hydrogen bonds, including:**
   a. **Relatively high melting and boiling points**
      More heat energy is required to break the relatively strong hydrogen bonds in water than other Group 16 hydrides
   b. **Relative density of solid and liquid states**
      In the liquid state, there are typically 1-2 hydrogen bonds per molecule. In the solid state, the molecules arrange themselves to have 4 hydrogen bonds per atom, leaving a more open structure.
   c. **Surface tension, viscosity, cohesive and adhesive forces**
      Surface Tension: The strong hydrogen bonds pull molecules on the surface towards the centre, so water is very resistant to increasing surface area
      Viscosity: Strong hydrogen bonds don't break easily, so it doesn't flow very easily
2. **Classify solvents as polar or non-polar, and solutes as ionic, polar and non-polar**

Water (Polar Solvent)	**C_6H_{12} (Non-Polar Solvent)**
CH_3COCH_3 (polar section)	C_6H_6 (non-polar)
CH_3COOH (polar; only small hydrophobic section)	CH_3COCH_3 (non-polar section; dissolves better in water)
NH_3 (polar)	
HCl (polar)	
NaCl (Ionic)	

   NOTE: Watch out for insoluble solvents
3. **Identify and describe interactions between solvent and solute, and use this to predict whether a solution will form**
   Polar solvent + ionic solute: Ion-Dipole forces - stronger than hydrogen bonds
   Polar solvent + polar solute: Dipole-dipole forces
   Non-polar solvent: Dispersion forces
   If the solvent-solute forces are strong enough to overcome the solute-solute and solvent-solvent forces the solution will form
4. **Describe and explain how ion-dipole forces form in solutions, and use this to explain why ionic solutes are often soluble in water or other polar solvents**
   Polar water molecules form electrostatic forces of attraction with the ions. These are quite strong (stronger than hydrogen bonds) and if enough water molecules form these bonds with an ion, they are enough to overcome the ionic forces holding the ion to its compound, and it is dissolved in the solution. As these are strong attractive forces, ionic solutes are generally soluble in polar solvents.
5. **Describe solutions using the terms saturated, unsaturated or supersaturated, and use these ideas to describe the concentration of a solution**
   Unsaturated: A solution in which more solute may be dissolved at a particular temperature
   Saturated: No more solute may be dissolved in the solution at a particular temperature
   Supersaturated: An unstable solution in which more solute is dissolved than is normally possible at a particular temperature. If disturbed, the solute will crystallise leaving a saturated solution
6. **Describe the concentration of a substance using moles per litre of solution (mol L^{-1}), mass per litre of solution (g L^{-1}) or parts per million (ppm)**
   Moles per litre (mol L^{-1}) $c = \frac{n}{V}$
   Grams per litre (g L^{-1}): $c = \frac{m}{V}$

Parts per million (ppm): $c = \frac{m\ (solute\ in\ mg)}{m\ (solvent\ in\ kg)} = \frac{m\ (solute\ in\ g)}{m\ (solvent\ in\ g)} * 1000\ 000$

7. **Convert concentrations between different units (listed above)**
8. **Describe and explain the relationship between solubility and temperature, for both solids and gases**

   An increase in temperature makes the particles of solute solvent have more kinetic energy:

   Solubility of solids *generally* increases (higher kinetic energy makes the bonds between them easier to break)

   Solubility of gases decreases (higher kinetic energy causes molecules to break intermolecular forces and escape the solution)
9. **Draw, interpret and analyse solubility graphs for solids and gases**
10. **Use understanding of solubility, along with the data tables, to predict whether solutions will form a precipitate**
11. **Write equations to show only the reacting species and products made in a reaction, including net ionic equations for precipitation reactions**

    Net ionic equation = remove spectator ions

    Always simplify coefficient ratio
12. **Give observations for reactions involving solutions, solids and precipitates, with colours of solutions and solids, using the Data Booklet**

    Solutions are always clear, but need to specify colour/colourless
13. **Conduct experimental work safely, competently and methodically to determine solubility of ionic compounds**
14. **State that potable drinking water must undergo various treatment processes to ensure it meets regulations for safe levels of solutes, including heavy metals**
    a. Aeration: Water sprayed into the air to reduce presence of dissolved gases such as CO_2
    b. Flocculation: Small suspended particles are made to clump together into larger particles which eventually sink to the bottom due to their weight. Achieved by adding alum ($Al_2(SO_4)_3$) and $Ca(OH)_2$ which precipitate $Al(OH)_3$, a gelatinous substance which suspended particles adsorb to
    c. Settling: Mixture left for some time to allow floc and gathered particles to sink to the bottom, then are removed
    d. Filtration: Water filtered through a bed of sand on gravel to remove any remaining suspended matter
    e. Chlorination/Clarification: Chlorine added to water to remove biological contaminants
    f. Fluoridation: Fluoride added (in the form of compounds) to increase fluorine concentration as it reacts with enamel to help prevent tooth decay
    g. CO_2 and $Ca(OH)_2$ (lime) added if necessary to bring pH between 6.5-8.5
15. **Explain why drinking water must be monitored for safe levels on solutes**

    Many solutes are hazardous to human health, such as pesticides from agricultural runoff, heavy metals from rainfall runoff on roads (zinc in car tyres/brake linings), and pathogens from animal waste. All of these must be removed to a certain level to ensure that the water will be safe to drink (potable)

Reactions Involving Acids/Bases

Equations of Acidic Solutions

- Acid + Reactive Metal → Salt + $H_{2(g)}$
- Acid + Metal Oxide/Hydroxide → Salt + $H_2O_{(l)}$
- Acid + Carbonate/Hydrogen carbonate → Salt + $H_2O_{(l)}$ + $CO_{2(g)}$
  - This is a two-stage process: Initially forms carbonic acid (H_2CO_3) which decomposes into $H_2O_{(l)}$ + $CO_{2(g)}$
- Acid + Metal Sulfite → Salt + $H_2O_{(l)}$ + $SO_{2(g)}$

Equations of Basic Solutions

- Base + Ammonium Salt → Salt + $H_2O_{(l)}$ + $NH_{3(g)}$
- Base + Non-Metal Oxide → Salt + $H_2O_{(l)}$